

# Projects Portfolio

Embedded Systems · Aerospace · Automotive · Cybersecurity · Medical

## E2-CAD

### EV Charging Station / V2G

2025 - 2026

Reverse engineering and analysis of embedded code on 3 boards (Microchip PIC, STM32). Test bench design and setup. Charging station debugging.

ISO 15118

OCPP

PLC

HomePlug

Linux

Python

#### Details

V2G (Vehicle-to-Grid) project. ISO 15118 and OCPP 1.6/2.0 protocols. PLC HomePlug Green PHY communication analysis between vehicle and charging station. Development of diagnostic tools in Python on Linux.

## SAFRAN

### SSPC A350 — Solid State Power Controller

2024 - 2025

Low-level driver development for the A350 program power controller. On-target validation testing.

C

Lauterbach TRACE32

DO-178

#### Details

DO-178C DAL-A context. C development on real-time microcontroller. Advanced debugging with Lauterbach TRACE32 probe. Unit and validation tests on target board.

## THALES

### Defense Radio LPM + N-MMR

2022 - 2023

Application layer development under embedded Linux for 2 military radio projects. HLT testing.

Linux

C/C++

Websockets

Bash

#### Details

Two defense radio projects: LPM (Military Programming Law) and N-MMR. C/C++ development on embedded Linux. Real-time Websockets communication. Bash scripts for HLT (High Level Tests) automation.

## VALEO

### Valeo Egypt Audit

2022

Audit of unit and integration testing processes on-site in Egypt. Improvement proposals.

NI Mastre

Vector CANoe

#### Details

Audit mission at the Valeo site in Cairo. Evaluation of test tools (NI Mastre, Vector CANoe). Quality process analysis and improvement proposals for validation teams.

**ZODIAC / SAFRAN**

## **GENOME / CORAC — SSPC A350**

2017 - 2020

Low-level drivers, uAFDX to CAN gateway, diagnostics. Flight test campaign at UBC Vancouver, Canada.

Microchip dsPIC

CAN

RS485

DO-178

### **Details**

CORAC/GENOME research program. Microchip dsPIC driver development. uAFDX to CAN bus communication gateway. RS485 protocol. Flight test campaign at the University of British Columbia (UBC), Vancouver.

**ENEDIS**

## **Linky Cybersecurity**

2015 - 2016

Penetration testing and security audit of Linky data concentrators. Intrusion tests on embedded Linux systems.

Kali Linux

Nmap

Wireshark

Scapy

SDR

Python

### **Details**

Penetration tests on Linky smart meter data concentrators. Network analysis with Nmap, Wireshark, Scapy. SDR radio. System security audit under Kali Linux. Vulnerability report.

**SORIN**

## **Platinum — BLE Pacemaker**

2014 - 2015

Feasibility study for Bluetooth Smart integration in a cardiac implant. Smartphone protocol.

Cortex M0

BLE

Android

IEC 62304

### **Details**

IEC 62304 context (medical software). Feasibility study for Bluetooth Low Energy integration in a pacemaker. Cortex M0 firmware development and Android communication application.

**ASSOCIATION**

## **Workshop-DIY**

2020 - present

Founder. 18+ workshops for youth. STEM education. Arduino, ESP32, drone, and AI projects.

Arduino

ESP32

AI

3D Printing

Robotics

### **Details**

Founder of the Workshop-DIY association. Organized over 18 educational workshops for young people (ages 8-18). Hands-on projects: Arduino, ESP32, 3D printing, drones, artificial intelligence, robotics.

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